BACHELOR OF COMPUTER APPLICATIONS (BCA)

SEMESTER VI - END SEMESTER EXAMINATION

December 2023

Paper Code: BCA-610

Subject: Artificial Intelligence and Machine Learning

Time Allowed: 3 Hours | Maximum Marks: 75

General Instructions:

1. Attempt all sections. Section A is compulsory; internal choice is given in Sections B and C.
2. Write answers in legible handwriting and number them exactly as in the question paper.
3. Draw neat, labelled diagrams wherever required.

SECTION A (5 x 3 = 15 Marks)

Answer all five questions. Each question carries 3 marks.

1. Define artificial intelligence. [3]

2. What is supervised learning? [3]

3. Explain the Turing test. [3]

4. What is overfitting? [3]

5. Define a neural network. [3]

SECTION B (4 x 10 = 40 Marks)

Answer any four questions. Each question carries 10 marks.

1. Explain the types of machine learning with examples. [10]

2. Explain linear regression with a simple example. [10]

3. Describe the perceptron model. [10]

4. Explain the evaluation metrics accuracy, precision and recall. [10]

SECTION C (2 x 10 = 20 Marks)

Answer any two questions. Each question carries 10 marks.

1. Explain the A* search algorithm with an example. [10]

2. Discuss the ethical issues and limitations of AI systems. [10]

*** *END OF PAPER* ***